

Inverse Functions (Part II)

Steps for Finding the Equation of an Inverse Function

I) Change $f(x)$ to y to make the equation easier to write.

II) Replace x 's with y 's and y 's with x 's.

III) Solve for y (i.e. isolate y).

IV) Change the new isolated y to $f^{-1}(x)$.

Find the equation of the inverse function.

• ① $f(x) = 3(4^x) - 9$

I) $y = 3(4^x) - 9$

II) $x = 3(4^y) - 9$

III) $x + 9 = 3(4^y)$

$$\frac{x+9}{3} = \frac{3(4^y)}{3}$$

$$\frac{x+9}{3} = 4^y$$

$$y = \log_4\left(\frac{x+9}{3}\right)$$

IV) $f^{-1}(x) = \log_4\left(\frac{x+9}{3}\right)$

② $f(x) = 2^{\frac{x}{3}+1}$

I) $y = 2^{\frac{x}{3}+1}$

II) $x = 2^{\frac{y}{3}+1}$

III) $\frac{y}{3} + 1 = \log_2 x$

$$\frac{y}{3} = \log_2 x - 1$$

$$3\left(\frac{y}{3}\right) = (\log_2 x - 1)(3)$$

$$y = 3\log_2 x - 3$$

IV) $f^{-1}(x) = 3\log_2 x - 3$

• ③ $f(x) = \frac{5^x}{4} + 1$

I) $y = \frac{5^x}{4} + 1$

II) $x = \frac{5^y}{4} + 1$

III) $x - 1 = \frac{5^y}{4}$

$$4(x-1) = \left(\frac{5^y}{4}\right)(4)$$

$$4x - 4 = 5^y$$

$$y = \log_5(4x - 4)$$

IV) $f^{-1}(x) = \log_5(4x - 4)$

④ $f(x) = 6^{-2x-5}$

I) $y = 6^{-2x-5}$

II) $x = 6^{-2y-5}$

III) $-2y - 5 = \log_6 x$

$$-2y = \log_6 x + 5$$

$$y = \frac{\log_6 x + 5}{-2}$$

IV) $f^{-1}(x) = -\frac{\log_6 x + 5}{2}$

• ⑤ $f(x) = \log_2(7x-1)$

I) $y = \log_2(7x-1)$

II) $x = \log_2(7y-1)$

III) $7y-1 = 2^x$

$$7y = 2^x + 1$$

$$y = \frac{2^x + 1}{7}$$

IV) $f^{-1}(x) = \frac{2^x + 1}{7}$

⑥ $f(x) = \frac{\log_4 x}{2} - 8$

I) $y = \frac{\log_4 x}{2} - 8$

II) $x = \frac{\log_4 y}{2} - 8$

III) $x+8 = \frac{\log_4 y}{2}$

$$2(x+8) = \log_4 y$$

$$2x+16 = \log_4 y$$

$$y = 4^{2x+16}$$

IV) $f^{-1}(x) = 4^{2x+16}$

• ⑦ $f(x) = 1 + \log_5(3x)$

I) $y = 1 + \log_5(3x)$

II) $x = 1 + \log_5(3y)$

III) $x-1 = \log_5(3y)$

$$3y = 5^{x-1}$$

$$y = \frac{5^{x-1}}{3}$$

IV) $f^{-1}(x) = \frac{5^{x-1}}{3}$

⑧ $f(x) = 5\log_6(x+4)$

I) $y = 5\log_6(x+4)$

II) $x = 5\log_6(y+4)$

III) $\frac{x}{5} = \log_6(y+4)$

$$y+4 = 6^{\frac{x}{5}}$$

$$y = 6^{\frac{x}{5}} - 4$$

IV) $f^{-1}(x) = 6^{\frac{x}{5}} - 4$

• ⑨ $f(x) = \frac{4}{5x}$

I) $y = \frac{4}{5x}$

II) $x = \frac{4}{5y}$

III) $(5y)x = \frac{4}{5y} (5y)$

$$5xy = 4$$

$$\frac{5xy}{5x} = \frac{4}{5x}$$

$$y = \frac{4}{5x}$$

IV) $f^{-1}(x) = \frac{4}{5x}$

• ⑩ $f(x) = \frac{1}{x}$

I) $y = \frac{1}{x}$

II) $x = \frac{1}{y}$

III) $(y)x = \frac{1}{y} (y)$

$$xy = 1$$

$$\frac{xy}{x} = \frac{1}{x}$$

$$y = \frac{1}{x}$$

IV) $f^{-1}(x) = \frac{1}{x}$

• ⑪ $f(x) = \frac{1}{x+6}$

I) $y = \frac{1}{x+6}$

II) $x = \frac{1}{y+6}$

III) $x(y+6) = \frac{1}{y+6} (y+6)$

$$xy + 6x = 1$$

$$xy = 1 - 6x$$

$$\frac{xy}{x} = \frac{1-6x}{x}$$

$$y = \frac{1-6x}{x}$$

IV) $f^{-1}(x) = \frac{1-6x}{x}$

• ⑫ $f(x) = \frac{2}{3x-1}$

I) $y = \frac{2}{3x-1}$

II) $x = \frac{2}{3y-1}$

III) $x(3y-1) = \frac{2}{3y-1} (3y-1)$

$$3xy - x = 2$$

$$3xy = 2 + x$$

$$\frac{3xy}{3x} = \frac{2+x}{3x}$$

$$y = \frac{2+x}{3x}$$

IV) $f^{-1}(x) = \frac{2+x}{3x}$

In the following problems, restrictions must be used on the original function and/or inverse function so that they are perfectly matched and no aberrations occur. To understand this, graph the function and note that it fails the Horizontal Line Test, which means that it is not one-to-one. Draw the inverse graph and note that it fails the Vertical Line Test. Highlight the parts of the inverse graph and the original graph that you will omit and use inequality notation next to the equations to indicate restrictions.

• ⑬ $f(x) = \cos x$, $0^\circ \leq x \leq 180^\circ$

I) $y = \cos x$

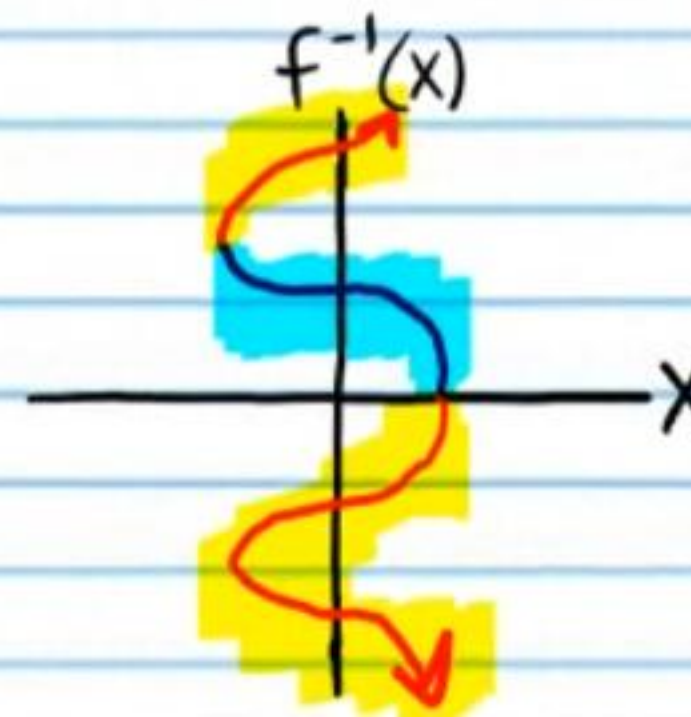
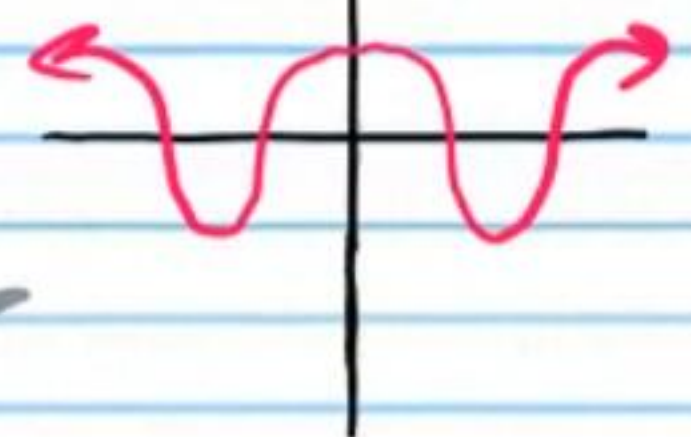
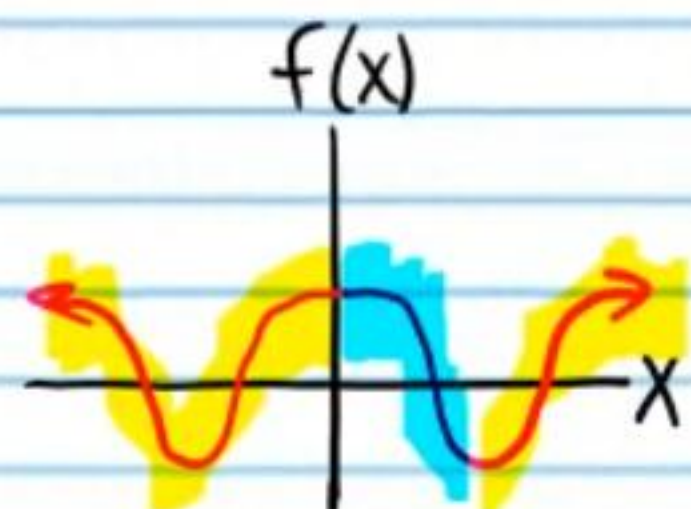
II) $x = \cos y$

III) $y = \cos^{-1} x$,

IV) $f^{-1}(x) = \cos^{-1} x$, $0^\circ \leq y \leq 180^\circ$

Reflect
across
the
y-axis.

Rotate
90°
Clockwise.



⑭ $f(x) = \csc x$, $-90^\circ \leq x \leq 90^\circ$, $x \neq 0^\circ$

I) $y = \csc x$

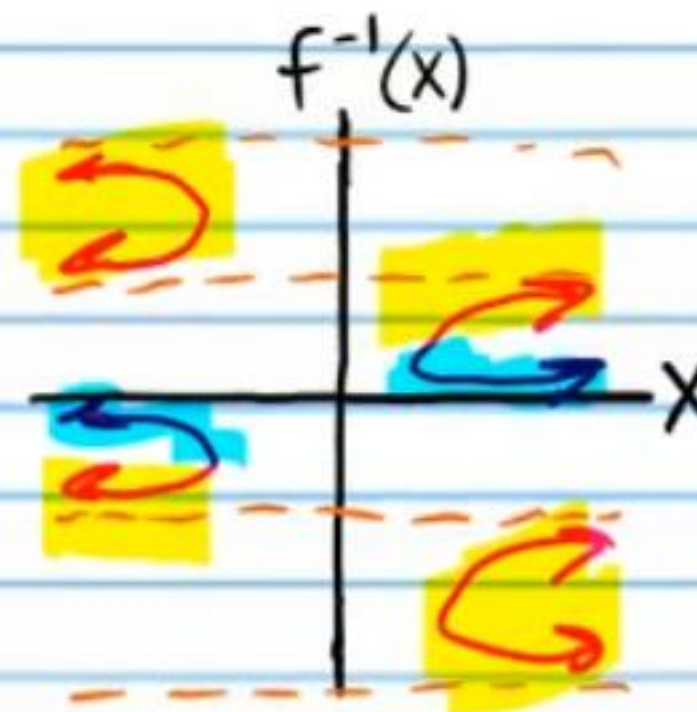
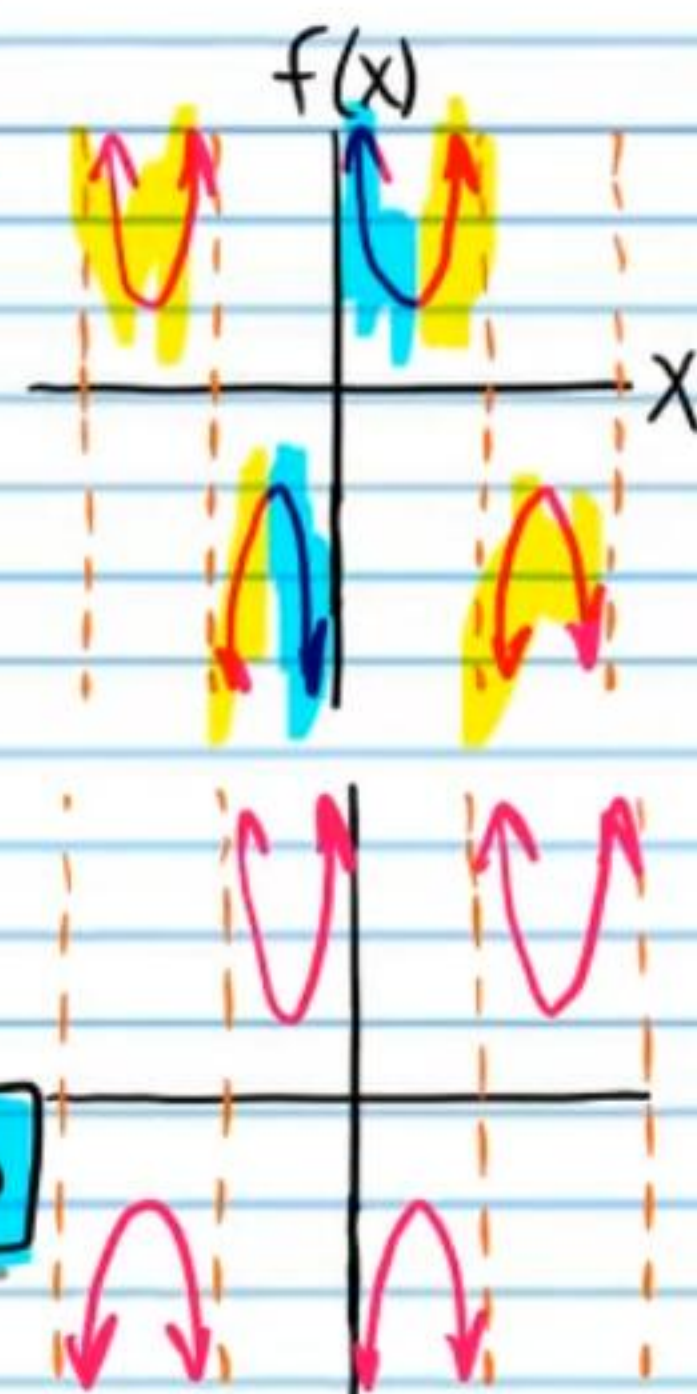
II) $x = \csc y$

III) $y = \csc^{-1} x$

IV) $f^{-1}(x) = \csc^{-1} x$, $-90^\circ \leq y \leq 90^\circ$, $y \neq 0$

Reflect
across
the
y-axis.

Rotate
90°
Clockwise.



• (15) $f(x) = \sin x$, $-90^\circ \leq x \leq 90^\circ$

I) $y = \sin x$

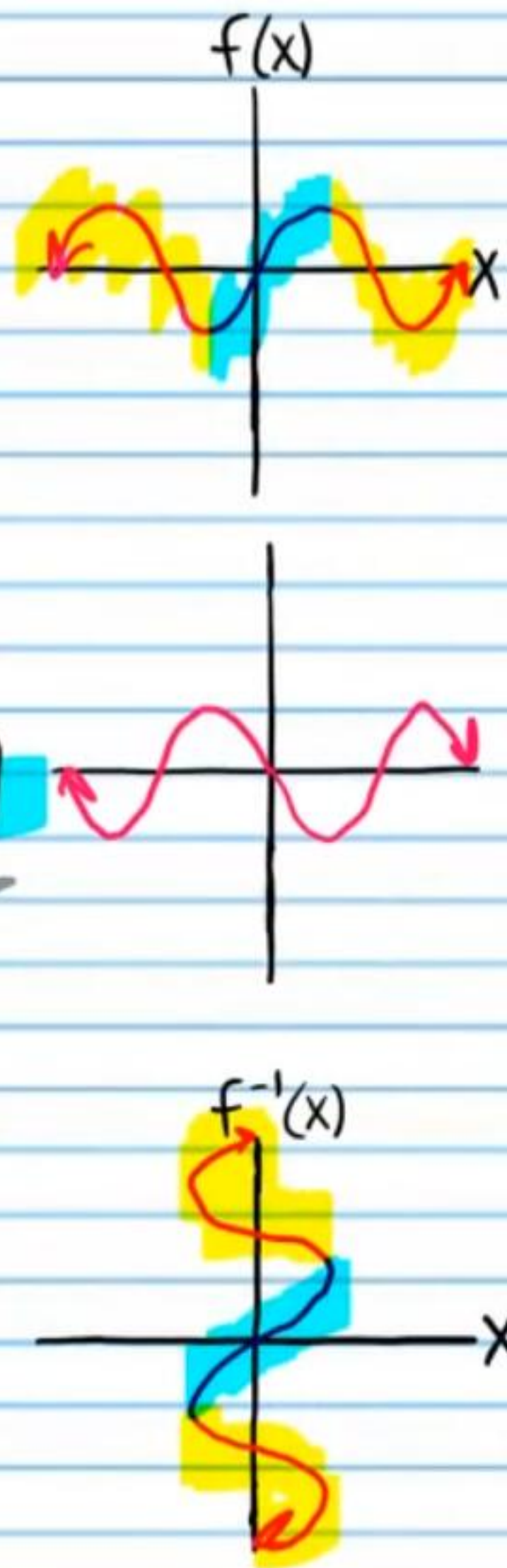
II) $x = \sin y$

III) $y = \sin^{-1} x$

IV) $f^{-1}(x) = \sin^{-1} x$, $-90^\circ \leq y \leq 90^\circ$

Reflect
across
the
y-axis.

Rotate
 90°
Clockwise.



(16) $f(x) = \tan x$, $-90^\circ < x < 90^\circ$

I) $y = \tan x$

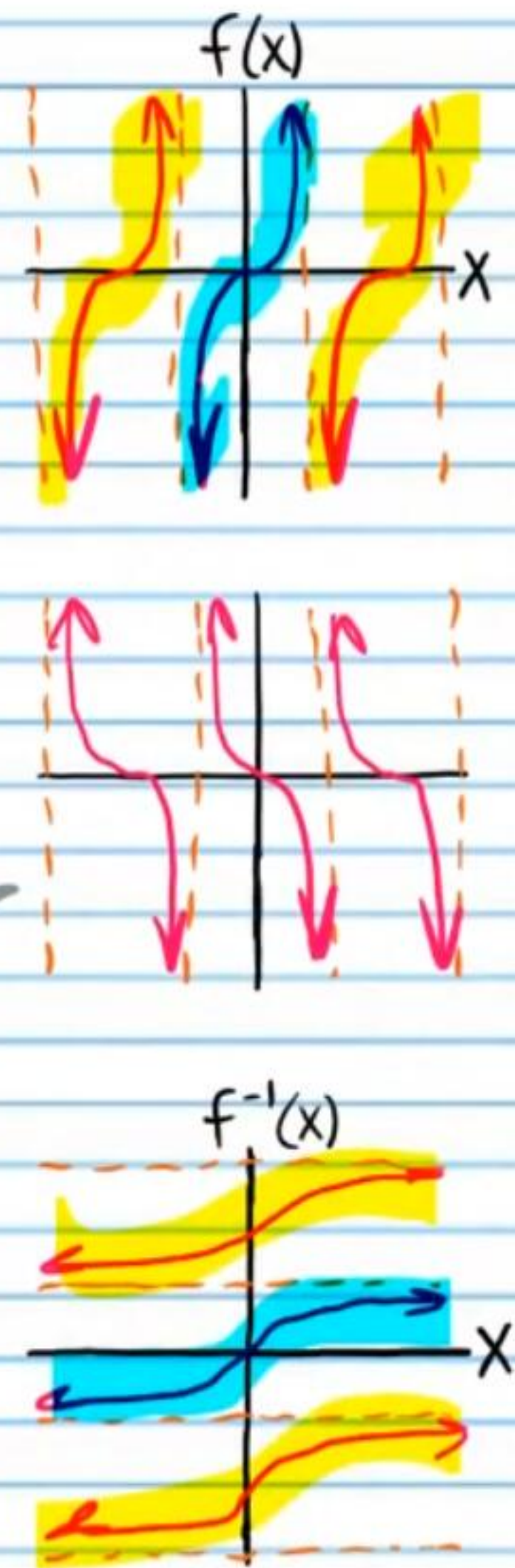
II) $x = \tan y$

III) $y = \tan^{-1} x$

IV) $f^{-1}(x) = \tan^{-1} x$, $-90^\circ < y < 90^\circ$

Reflect
across
the
y-axis.

Rotate
 90°
Clockwise.



Using Composition to Verify That Functions Are Inverses

$$(f \circ g)(x) = f(g(x)) \quad (g \circ f)(x) = g(f(x))$$

Verifying That Functions Are Inverses

If $f(g(x)) = x$ and $g(f(x)) = x$, then "f" and "g" are inverse functions. If "f" and "g" are inverse functions, then "g" can be named f^{-1} and/or "f" can be named g^{-1} .

Example: Let $f(x) = 2x + 3$ & $g(x) = \frac{x-3}{2}$.

$f(g(x))$ tells us to start with x , move forward, then reverse, arriving at x , where we started.

Start at x . Forward Reverse Back to start

$$f(g(x)) = 2\left(\frac{x-3}{2}\right) + 3 = \cancel{2}\left(\frac{x-3}{\cancel{2}}\right) + 3 = x - 3 + 3 = x$$

$g(f(x))$ tells us to start with x , move forward, then reverse, arriving at x , where we started.

Start at x . Forward Reverse Back to start.

$$g(f(x)) = \frac{2x+3-3}{2} = \frac{2x+\cancel{3}-3}{2} = \frac{2x}{2} = x$$

Show algebraically that "f" and "g" are inverse functions.

• (17) $f(x) = 6(x+10)+7$ & $g(x) = \frac{x-67}{6}$

$$f(g(x))$$

$$6(g(x)+10)+7$$

$$6\left(\frac{x-67}{6}+10\right)+7$$

$$6\left(\frac{x-67}{6}+\frac{60}{6}\right)+7$$

$$6\left(\frac{x-67+60}{6}\right)+7$$

$$6\left(\frac{x-7}{6}\right)+7$$

$$x-7+7$$

$$x$$

$$g(f(x))$$

$$\frac{f(x)-67}{6}$$

$$\frac{6(x+10)+7-67}{6}$$

$$\frac{6(x+10)-60}{6}$$

$$\frac{6x+60-60}{6}$$

$$\frac{6x}{6}$$

$$x$$

⑮ $f(x) = x^5 - 8$ & $g(x) = \sqrt[5]{x+8}$

$$f(g(x))$$

$$[g(x)]^5 - 8$$

$$[\sqrt[5]{x+8}]^5 - 8$$

$$x+8-8$$

$$x$$

$$g(f(x))$$

$$\sqrt[5]{f(x)+8}$$

$$\sqrt[5]{x^5-8+8}$$

$$\sqrt[5]{x^5}$$

$$x$$

• ⑯ $f(x) = \frac{x}{2} + 10$ & $g(x) = 2x - 20$

$$f(g(x))$$

$$\frac{g(x)}{2} + 10$$

$$\frac{2x-20}{2} + 10$$

$$\frac{2x}{2} - \frac{20}{2} + 10$$

$$x - 10 + 10$$

$$x$$

$$g(f(x))$$

$$2[f(x)] - 20$$

$$2\left(\frac{x}{2} + 10\right) - 20$$

$$\frac{2x}{2} + 20 - 20$$

$$x + 20 - 20$$

$$x$$

② $f(x) = 2x^5 - 1$ & $g(x) = \sqrt[5]{\frac{x+1}{2}}$

$f(g(x))$

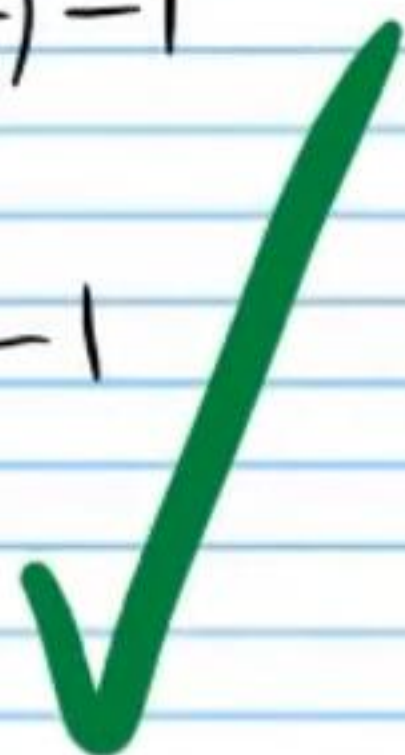
$2[g(x)]^5 - 1$

$2\left(\sqrt[5]{\frac{x+1}{2}}\right)^5 - 1$

$2\left(\frac{x+1}{2}\right) - 1$

$x+1-1$

x



$g(f(x))$

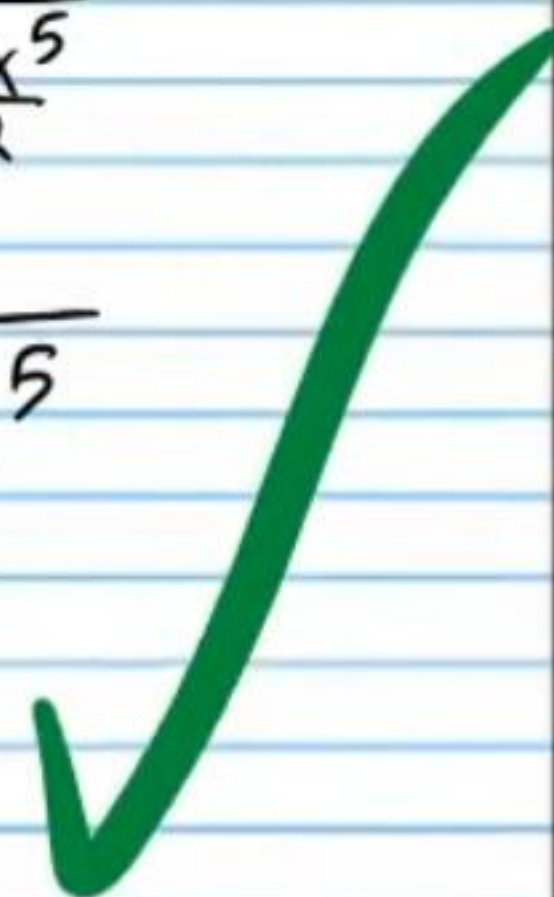
$\sqrt[5]{\frac{f(x)+1}{2}}$

$\sqrt[5]{\frac{2x^5-1+1}{2}}$

$\sqrt[5]{\frac{2x^5}{2}}$

$\sqrt[5]{x^5}$

x



Logarithm Property I

Logarithmic equations and exponential equations are equivalent. In other words, $x = \log_b C$ if and only if $C = b^x$. We learned in geometry class that "if and only if" means the converse of the statement is true also. Note: Traditionally, neither "C" nor "b" is allowed to be negative or zero when placed in a logarithmic expression.

Logarithm Property II

$\log_b b^x = x$

Logarithm Property III

$\log_b b = 1$

Logarithm Property IV

$b^{\log_b x} = x$

Logarithm Property V

$\log_k(AB) = \log_k A + \log_k B$

Logarithm Property VI

$\log_k\left(\frac{A}{B}\right) = \log_k A - \log_k B$

Logarithm Property VII

$\log_k A^B = B \log_k A$

Logarithm Property VIII

If $\log_k A = \log_k B$, then $A = B$.

Show algebraically that "f" and "f⁻¹" are inverse functions.

• (21) $f(x) = 2^x + 1$ & $f^{-1}(x) = \log_2(x-1)$

$$f(f^{-1}(x))$$

$$2^{f^{-1}(x)} + 1$$

$$2^{\log_2(x-1)} + 1$$

$$x-1+1$$

$$x$$

$$f^{-1}(f(x))$$

$$\log_2(f(x)-1)$$

$$\log_2(2^x+1-1)$$

$$\log_2 2^x$$

$$x$$

(22) $f(x) = \log_2(7x-1)$ & $f^{-1}(x) = \frac{2^x+1}{7}$

$$f(f^{-1}(x))$$

$$\log_2[7(f^{-1}(x))-1]$$

$$\log_2[7(\frac{2^x+1}{7})-1]$$

$$\log_2(2^x+1-1)$$

$$\log_2 2^x$$

$$x$$

$$f^{-1}(f(x))$$

$$\frac{2^{f(x)}+1}{7}$$

$$\frac{2^{\log_2(7x-1)}+1}{7}$$

$$\frac{7x-1+1}{7}$$

$$\frac{7x}{7}$$

$$x$$

• ②③ $f(x) = 4(3^x) - 1$ & $f^{-1}(x) = \log_3\left(\frac{x+1}{4}\right)$

$$f(f^{-1}(x))$$

$$4(3^{f^{-1}(x)}) - 1$$

$$4(3^{\log_3(\frac{x+1}{4})}) - 1$$

$$4\left(\frac{x+1}{4}\right) - 1$$

$$x+1-1$$

$$x$$

$$f^{-1}(f(x))$$

$$\log_3\left(\frac{f(x)+1}{4}\right)$$

$$\log_3\left(\frac{4(3^x)-1+1}{4}\right)$$

$$\log_3\left(\frac{4(3^x)}{4}\right)$$

$$\log_3 3^x$$

$$x$$

②④ $f(x) = 2\log_4 x - 8$ & $f^{-1}(x) = 4^{\left(\frac{x+8}{2}\right)}$

$$f(f^{-1}(x))$$

$$2\log_4[f^{-1}(x)] - 8$$

$$2\log_4 4^{\left(\frac{x+8}{2}\right)} - 8$$

$$2\left(\frac{x+8}{2}\right) - 8$$

$$x+8-8$$

$$x$$

$$f^{-1}(f(x))$$

$$4^{\left(\frac{f(x)+8}{2}\right)}$$

$$4^{\left(\frac{2\log_4 x - 8 + 8}{2}\right)}$$

$$4^{\frac{2\log_4 x}{2}}$$

$$4^{\log_4 x}$$

$$x$$

• (25) $f(x) = \frac{4}{5x}$ & $f^{-1}(x) = \frac{4}{5x}$

$$f(f^{-1}(x))$$

$$\frac{4}{5[f^{-1}(x)]}$$

$$\frac{4}{5(\frac{4}{5x})}$$

$$\frac{4}{\frac{4}{x}}$$

$$4 \div \frac{4}{x}$$

$$\frac{4}{1} \div \frac{4}{x}$$

$$\frac{4}{1} \left(\frac{x}{4} \right)$$

$$\frac{4x}{4}$$

$$x$$

$$f^{-1}(f(x))$$

$$\frac{4}{5(f(x))}$$

$$\frac{4}{5(\frac{4}{5x})}$$

$$\frac{4}{\frac{4}{x}}$$

$$4 \div \frac{4}{x}$$

$$\frac{4}{1} \div \frac{4}{x}$$

$$\frac{4}{1} \left(\frac{x}{4} \right)$$

$$\frac{4x}{4}$$

$$x$$

• (26) $f(x) = \frac{1}{x}$ & $f^{-1}(x) = \frac{1}{x}$

$$f(f^{-1}(x))$$

$$\frac{1}{f^{-1}(x)}$$

$$\frac{1}{\frac{1}{x}}$$

$$1 \div \frac{1}{x}$$

$$\frac{1}{1} \div \frac{1}{x}$$

$$\frac{1}{1} \left(\frac{x}{1} \right)$$

$$\frac{x}{1}$$

$$x$$

$$f^{-1}(f(x))$$

$$\frac{1}{f(x)}$$

$$\frac{1}{\frac{1}{x}}$$

$$1 \div \frac{1}{x}$$

$$\frac{1}{1} \div \frac{1}{x}$$

$$\frac{1}{1} \left(\frac{x}{1} \right)$$

$$\frac{x}{1}$$

$$x$$

• (27) $f(x) = \frac{2}{3x-1}$ & $f^{-1}(x) = \frac{2+x}{3x}$

$$f(f^{-1}(x))$$

$$\frac{2}{3[f^{-1}(x)]-1}$$

$$\frac{2}{3\left(\frac{2+x}{3x}\right)-1}$$

$$\frac{2}{\frac{2+x}{x}-1}$$

$$\frac{2}{\frac{2+x}{x}-\frac{x}{x}}$$

$$\frac{2}{2+x-x}$$

$$\frac{2}{\frac{2}{x}}$$

$$\frac{2}{\frac{2}{x}}$$

$$2 \div \frac{2}{x}$$

$$2\left(\frac{x}{2}\right)$$

x

$$f^{-1}(f(x))$$

$$\frac{2+f(x)}{3f(x)}$$

$$\frac{2+\frac{2}{3x-1}}{3\left(\frac{2}{3x-1}\right)}$$

$$\frac{\frac{2}{1} \frac{(3x-1)}{(3x-1)} + \frac{2}{3x-1}}{\frac{6}{3x-1}}$$

$$\frac{\frac{6x-2}{3x-1} + \frac{2}{3x-1}}{\frac{6}{3x-1}}$$

$$\frac{\frac{6x-2+2}{3x-1}}{\frac{6}{3x-1}}$$

$$\frac{\frac{6x-2+2}{3x-1}}{\frac{6}{3x-1}}$$

$$\frac{\frac{6x}{3x-1}}{\frac{6}{3x-1}}$$

$$\frac{6x}{3x-1} \div \frac{6}{3x-1}$$

$$\frac{6x}{3x-1} \cdot \frac{3x-1}{6}$$

x

• 28) $f(x) = \frac{1}{x+6}$ & $f^{-1}(x) = \frac{1-6x}{x}$

$$f(f^{-1}(x))$$

$$\frac{1}{f^{-1}(x)+6}$$

$$\frac{1}{\frac{1-6x}{x}+6}$$

$$\frac{1}{\frac{1-6x}{x} + \frac{6x}{x}}$$

$$\frac{1}{\frac{1-6x+6x}{x}}$$

$$\frac{1}{\frac{1}{x}}$$

$$\frac{1}{x}$$

$$1 \div \frac{1}{x}$$

$$\frac{1}{1} \div \frac{1}{x}$$

$$\frac{1}{1} \cdot \frac{x}{1}$$

$$\frac{x}{1}$$

$$x$$

$$f^{-1}(f(x))$$

$$\frac{1-6f(x)}{f(x)}$$

$$\frac{1-6\left(\frac{1}{x+6}\right)}{\frac{1}{x+6}}$$

$$\frac{1-\frac{6}{x+6}}{\frac{1}{x+6}}$$

$$\frac{\frac{1}{1}-\frac{6}{x+6}}{\frac{1}{x+6}}$$

$$\frac{\frac{1}{1}(x+6)-\frac{6}{x+6}}{\frac{1}{x+6}}$$

$$\frac{\frac{x+6}{x+6}-\frac{6}{x+6}}{\frac{1}{x+6}}$$

$$\frac{\frac{x+6-6}{x+6}}{\frac{1}{x+6}}$$

$$\frac{\frac{x}{x+6}}{\frac{1}{x+6}}$$

$$\frac{x}{x+6} \div \frac{1}{x+6}$$

$$\frac{x}{x+6} \cdot \frac{x+6}{1}$$

$$\frac{x}{1}$$

$$x$$